

Advanced Calculus HW

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Question 1

(a)

$\vec{r}(t) = (\cosh(t), \sinh(t), 3)$ and we need

$$\vec{T} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}, \quad \vec{N} = \frac{\vec{T}'(t)}{|\vec{T}'(t)|}, \quad \vec{B} = \vec{T} \times \vec{N}$$

Then

$$\begin{aligned}\vec{r}'(t) &= (\sinh(t), \cosh(t), 0) \\ |\vec{r}'(t)| &= \sqrt{\sinh^2(t) + \cosh^2(t)}\end{aligned}$$

Thus

$$\vec{T}(t) = \frac{1}{\sqrt{\sinh^2(t) + \cosh^2(t)}}(\sinh(t), \cosh(t), 0)$$

Now calculating $\vec{T}'(t)$,

(simplifications $\cosh^2(t) + \sinh^2(t) = \cosh(2t)$, $\sinh(2t) = 2 \sinh(t) \cosh(t)$ and $\cosh^2(t) - \sinh^2(t) = 1$ used)

$$\begin{aligned}\vec{T}'(t) &= \frac{1}{\sqrt{\cosh(2t)}}(\cosh(t), \sinh(t), 0) - \frac{\sinh(2t)}{(\cosh(2t))^{\frac{3}{2}}}(\sinh(t), \cosh(t), 0) \\ &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}}(\cosh(2t)(\cosh(t), \sinh(t), 0) - (2 \sinh(t) \cosh(t))(\sinh(t), \cosh(t), 0)) \\ &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}}(\cosh(t) \cosh(2t) - 2 \sinh^2(t) \cosh(t), \sinh(t) \cosh(2t) - 2 \sinh(t) \cosh^2(t), 0) \\ &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}}(\cosh^3(t) + \sinh^2(t) \cosh(t) - 2 \sinh^2(t) \cosh(t), \sinh^3(t) + \cosh^2(t) \sinh(t) - 2 \sinh(t) \cosh^2(t), 0) \\ &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}}(\cosh(t)(\cosh^2(t) - \sinh^2(t)), \sinh(t)(\sinh^2(t) - \cosh^2(t)), 0) \\ &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}}(\cosh(t)(1), \sinh(t)(-1), 0) \\ &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}}(\cosh(t), -\sinh(t), 0)\end{aligned}$$

$$\begin{aligned}|\vec{T}'(t)| &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}} \sqrt{\cosh^2(t) + \sinh^2(t)} \\ &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}} \sqrt{\cosh(2t)} \\ &= \frac{1}{\cosh(2t)}\end{aligned}$$

$$\begin{aligned}\vec{N}(t) &= \frac{1}{(\cosh(2t))^{\frac{3}{2}}}(\cosh(t), -\sinh(t), 0)(\cosh(2t)) \\ &= \frac{1}{\sqrt{\cosh(2t)}}(\cosh(t), -\sinh(t), 0)\end{aligned}$$

Finally calculating $\vec{B}(t)$

$$\begin{aligned}\vec{B}(t) &= \left(\frac{1}{\sqrt{\cosh(2t)}}(\sinh(t), \cosh(t), 0) \right) \times \left(\frac{1}{\sqrt{\cosh(2t)}}(\cosh(t), -\sinh(t), 0) \right) \\ &= \left(\frac{1}{\sqrt{\cosh(2t)}} \right)^2 \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \sinh(t) & \cosh(t) & 0 \\ \cosh(t) & -\sinh(t) & 0 \end{bmatrix} \\ &= \frac{1}{\cosh(2t)}(0, 0, -\sinh^2(t) - \cosh^2(t)) \\ &= \frac{1}{\cosh(2t)}(0, 0, -\cosh(2t)) \\ &= (0, 0, -1)\end{aligned}$$

(b)

For an arbitrary smooth parameter t , $\kappa(t)$ is given by $\frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}$

$$\begin{aligned}\vec{r}'(t) &= (\sinh(t), \cosh(t), 0) \\ \vec{r}''(t) &= (\cosh(t), \sinh(t), 0)\end{aligned}$$

$$\begin{aligned}\vec{r}'(t) \times \vec{r}''(t) &= \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \sinh(t) & \cosh(t) & 0 \\ \cosh(t) & \sinh(t) & 0 \end{bmatrix} \\ &= (0, 0, \sinh^2(t) - \cosh^2(t)) \\ &= (0, 0, -1)\end{aligned}$$

$$|\vec{r}'(t) \times \vec{r}''(t)| = 1$$

$$|\vec{r}'(t)| = \sqrt{\cosh(2t)}$$

Thus,

$$\kappa(t) = \frac{1}{(\cosh(2t))^{\frac{3}{2}}}$$

Then for the points corresponding to $t = 0$ and $t = \frac{\ln(2)}{2}$, we get

$$\kappa(0) = \frac{1}{1} = 1, \quad \kappa\left(\frac{\ln(2)}{2}\right) = \frac{1}{\left(\frac{5}{4}\right)^{\frac{3}{2}}} = \frac{8}{5\sqrt{5}}$$

Now finding the tangent line to the curve at the point corresponding to $t = \frac{\ln(2)}{2}$,

The point is $(\cosh(\frac{\ln(2)}{2}), \sinh(\frac{\ln(2)}{2}), 3) = (\frac{3}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, 3)$, and points in the direction of $\vec{r}'(t)$, which is $(\frac{1}{2\sqrt{2}}, \frac{3}{2\sqrt{2}}, 0)$

Thus the vector equation of the line is

$$\vec{L} = \left(\frac{3}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, 3 \right) + s \left(\frac{1}{2\sqrt{2}}, \frac{3}{2\sqrt{2}}, 0 \right) \quad \forall s \in \mathbb{R}$$

Now we have to find the circle with the same tangent line and containing the point

It should have radius $\frac{1}{\kappa(t)} = \frac{5\sqrt{5}}{8}$

The centre of this circle should be one radius from the point in the direction of $\vec{N}(\frac{\ln(2)}{2}) = (\frac{3}{\sqrt{10}}, -\frac{1}{\sqrt{10}}, 0)$

$$C = \left(\frac{3}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}, 3 \right) + \frac{5\sqrt{5}}{8} \left(\frac{3}{\sqrt{10}}, -\frac{1}{\sqrt{10}}, 0 \right)$$

Then simplifying

$$C = \left(\frac{27}{8\sqrt{2}}, -\frac{1}{8\sqrt{2}}, 3 \right)$$

We now have the centre and radius of the circle meaning its equation is

$$\left(x - \frac{27}{8\sqrt{2}} \right)^2 + \left(y + \frac{1}{8\sqrt{2}} \right)^2 = \left(\frac{5\sqrt{5}}{8} \right)^2, \quad z = 3$$

Question 2

(a)

$$\vec{r}'(s) = \left(\frac{1}{2} \cos \left(1 + \frac{s}{2} \right), -\frac{1}{2} \sin \left(1 + \frac{s}{2} \right), \frac{\sqrt{3}}{2} \right)$$

$$|\vec{r}'(s)| = \sqrt{\frac{1}{4}(\cos^2 \left(1 + \frac{s}{2} \right) + \sin^2 \left(1 + \frac{s}{2} \right)) + \frac{3}{4}} = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$$

(b)

$$\kappa(s) = \left| \frac{d\vec{T}}{ds} \right| = |\vec{r}''(s)|$$

$$\vec{r}''(s) = \left(-\frac{1}{4} \sin \left(1 + \frac{s}{2} \right), -\frac{1}{4} \cos \left(1 + \frac{s}{2} \right), 0 \right)$$

$$|\vec{r}''(s)| = \sqrt{\frac{1}{16}} = \frac{1}{4}$$

Thus,

$$\kappa(s) = \frac{1}{4}$$

(c)

In arc length parameterizations we know $\vec{T} = \vec{r}'$, $\vec{N} = \frac{\vec{r}''}{|\vec{r}''|}$, $\vec{B} = \frac{\vec{r}' \times \vec{r}''}{|\vec{r}' \times \vec{r}''|}$

Thus

$$\vec{T}(s) = \left(\frac{1}{2} \cos \left(1 + \frac{s}{2} \right), -\frac{1}{2} \sin \left(1 + \frac{s}{2} \right), \frac{\sqrt{3}}{2} \right)$$

Then

$$\vec{N}(s) = \left(-\sin \left(1 + \frac{s}{2} \right), -\cos \left(1 + \frac{s}{2} \right), 0 \right)$$

Finally

$$\begin{aligned} \vec{B}(s) &= 4 \left(\left(\frac{1}{2} \cos \left(1 + \frac{s}{2} \right), -\frac{1}{2} \sin \left(1 + \frac{s}{2} \right), \frac{\sqrt{3}}{2} \right) \times \left(-\frac{1}{4} \sin \left(1 + \frac{s}{2} \right), -\frac{1}{4} \cos \left(1 + \frac{s}{2} \right), 0 \right) \right) \\ &= \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 \cos \left(1 + \frac{s}{2} \right) & -2 \sin \left(1 + \frac{s}{2} \right) & \frac{\sqrt{3}}{2} \\ -\sin \left(1 + \frac{s}{2} \right) & -\cos \left(1 + \frac{s}{2} \right) & 0 \end{bmatrix} \\ &= \left(\frac{\sqrt{3}}{2} \cos \left(1 + \frac{s}{2} \right), \frac{\sqrt{3}}{2} \sin \left(1 + \frac{s}{2} \right), -2 \cos^2 \left(1 + \frac{s}{2} \right) - 2 \sin^2 \left(1 + \frac{s}{2} \right) \right) \\ &= \left(\frac{\sqrt{3}}{2} \cos \left(1 + \frac{s}{2} \right), \frac{\sqrt{3}}{2} \sin \left(1 + \frac{s}{2} \right), -2 \right) \end{aligned}$$

(d)

$$\begin{aligned}\vec{B}'(s) &= \left(-\frac{\sqrt{3}}{4} \sin\left(1 + \frac{s}{2}\right), \frac{\sqrt{3}}{4} \cos\left(1 + \frac{s}{2}\right), 0 \right) \\ &= \frac{\sqrt{3}}{4} \vec{N}(s)\end{aligned}$$